

Fast Matrix Multiplication Algorithms In Rust

Computational Laboratory Project Report

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Abstract

We develop a Rust crate (library) that given matrices $A, B \in \mathbb{R}^{N \times N}$ computes their product $C = AB$ using fast matrix multiplication algorithms, for any given CP decomposition.

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1 Introduction

Matrix multiplication is one of the most fundamental computations in numerical linear algebra and scientific computing. We define *fast multiplication algorithms* as those that perform asymptotically fewer floating-point operations, and require asymptotically less data communication than the classical algorithm. In this work, we demonstrate that fast matrix multiplication algorithms are sensitive to the choice of base case, and perform better than their base case for larger sizes. We also provide a library function written in Rust to perform matrix multiplication with these algorithms, which supports parallel execution and different general matrix matrix multiplication *GEMM* libraries, and we perform our benchmarks with Intel Math Kernel Library (MKL) *dgemm* (double-precision general matrix-matrix multiplication) and *faer* (a modern, high performance linear algebra library for the Rust programming language), to evaluate performance of small matrix multiplication libraries integrated in *faer* scientific computing library, which are commonly used among users, as they work out-the-box compared to optimized GEMM libraries such as DGEMM, which require some setup. According to Benson and Ballard [1], fast matrix multiplication algorithms have largely been ignored by numerical libraries due to unfounded reasons. As they state, Intel’s MKL, AMD’s Core Math Library (ACML) do not provide implementations of fast algorithms. In reality, fast matrix multiplication algorithms perform fewer recursive matrix multiplications at the expense of more matrix additions, and are faster than classical ones both in theory and in practice.

Strassen’s algorithm is the best-known known fast algorithm, and will be at the core of our experiments, but we also support other algorithms among those listed in Table 2. We review those algorithms and methods to construct them in Section 2. In Section 3 we explain the technical details of our implementation, and in Section 4 we explain parallel and algorithmic optimizations. Finally, we summarize experimental results in Section 5. We summarize our findings as follows:

- Performance of base GEMM impacts performance of fast matrix multiplication algorithm,
- We evaluate Rust as a language in scientific computing, a domain where rust is in the early stages of adoption [2].

1.1 Notation and tensor preliminaries

The relevant notation for our work is in Table 1. We briefly review basic tensor preliminaries, following the notation in [3]. A *tensor* is a multidimensional array, and in this work we deal only with 3-dimensional (or 3-way) tensors $\mathcal{T} \in \mathbb{R}^{M \times N \times P}$. The *k-th frontal slice* of $\mathcal{T}$ is the matrix $\mathbf{X}_k$. For $\mathbf{u} \in \mathbb{R}^M$, $\mathbf{v} \in \mathbb{R}^N$, $\mathbf{w} \in \mathbb{R}^P$, we define the rank-1 *outer product* tensor $\mathcal{T} = \mathbf{u} \circ \mathbf{v} \circ \mathbf{w} \in \mathbb{R}^{M \times N \times P}$ with entries $x_{ijk} = u_i v_j w_k$; such tensor has *rank* one. Tensor addition is defined entry-wise. The rank of a tensor $\mathcal{T}$ is defined as the minimum number of rank-one tensors that generate $\mathcal{T}$ as their sum. We define the *CP decomposition* of rank R of a tensor $\mathcal{T}$ as the sum $\mathcal{T} = \sum_{r=1}^R u_r \circ v_r \circ w_r$, and we denote it by $\mathcal{T} = \llbracket U, V, W \rrbracket$, where U, V and W are matrices with R columns given by u_r, v_r , and w_r . An important operation in the context of tensors is the *mode- k product* between $\mathcal{T} \in \mathbb{R}^{N_1, N_2, N_3}$ and a matrix $U \in \mathbb{R}^{N_k \times R}$, for $k \in \{1, 2, 3\}$, which is defined as a tensor $\mathcal{Y} = \mathcal{X} \times_k U \in \mathbb{R}^{n_{k-1} \times R \times n_{k+1}}$ with $\mathbf{Y}_{(k)} = U \mathbf{X}_{(k)}$, also in components as $\mathcal{Y}(i_{k-1}, \alpha, i_{k+1}) = \sum_{i_k=1}^3 \mathcal{X}_{i_1 i_2 i_3} U_{\alpha, i_k}$.

Table 1: Summary of notation.

$\langle m, n, p \rangle$	Bilinear form representing base case matrix multiplication of an $m \times n$ matrix by a $n \times p$ matrix.
$M \times N \times P$	Dimensions of actual matrices that are multiplied ($M \times N$ matrix multiplied by $N \times P$ matrix).
$\llbracket \mathbf{U}, \mathbf{V}, \mathbf{W} \rrbracket$	Factor matrices corresponding to a tensor CP decomposition that provides a fast algorithm.
$\mathcal{T}$	Order-3 tensor.
$\mathcal{T} = \mathbf{u} \circ \mathbf{v} \circ \mathbf{w}$	Rank-1 tensor with entries $X_{ijk} = u_i v_j w_k$.
$\mathbf{T}_{(i)}$	i th frontal slice of $\mathcal{T}$, the matrix of entries $X_{:, :, i}$.
$A_{:,k}, A_{k:,} A_{ij}$	k th column, k th row and i, j entry of matrix $\mathbf{A}$.
$\text{vec}(\mathbf{A})$	Column-order vectorization of the entries of $\mathbf{A}$.
$\text{nnz}(\cdot)$	Number of non-zero entries of an object.

2 Fast matrix multiplication

2.1 Evaluation of bilinear form using low-rank tensor decompositions

Let X, Y, Z be finite-dimensional vector spaces of dimensions I, J, K respectively. A bilinear form is a mapping $f : X \times Y \rightarrow Z$ that is linear in each entry, in other words such that $f(x, \cdot)$ and $f(\cdot, y)$ are linear for all $x \in X$ and $y \in Y$. Each component f_k is a bilinear form $f_k : X \times Y \rightarrow \mathbb{R}$, which can be represented by a matrix slice $\mathbf{T}_{(k)}$ of a tensor $\mathcal{T}$ such as

$$(1) \quad f_k(x, y) = x^\top \mathbf{T}_{(k)} y = \sum_{i=1}^I \sum_{j=1}^J t_{ijk} x_i y_j$$

, for all $k \in \{1, \dots, K\}$. Hence we can construct a tensor $\mathcal{T}$ which represents f by taking the the slices $\mathbf{T}_{(k)}$ that represent f_k . In more succinct tensor notation,

$$(2) \quad f = \mathcal{T} \times_1 x^\top \times_2 y^\top.$$

Numerically evaluating the bilinear form in Equation 1 has a high computational cost driven by the multiplication that depends by the non-zero entries of $\mathcal{T}$. We call those multiplications *active* to emphasize their dominant computational role. We can reduce the cost in evaluating $f(x, y)$ by considering a CP decomposition $\llbracket \mathbf{U}, \mathbf{V}, \mathbf{W} \rrbracket$ of rank R of $\mathcal{T}$. Denoting $\mathbf{u}_1, \mathbf{v}_1, \mathbf{w}_1$ as the columns of $\mathbf{U}, \mathbf{V}$ and $\mathbf{W}$, we have the decomposition $\mathcal{T} = \sum_{r=1}^R u_r \circ v_r \circ w_r$, and by performing a substitution into Equation 2 component-wise, we get

(3)

$$\begin{aligned}
f_k(x, y) &= \left(\left(\sum_{r=1}^R u_l \circ v_r \circ w_r \right) \times_1 x^\top \times_2 y^\top \right)_k = \left(\sum_{r=1}^R (u_r \circ v_r \circ w_r) \times_1 x^\top \times_2 y^\top \right)_k \\
&= \sum_{i=1}^I \sum_{j=1}^J \sum_{r=1}^R u_{ir} v_{jr} w_{kr} x_i y_j = \sum_{r=1}^R \left(\sum_{i=1}^I u_{ir} x_i \right) \cdot \left(\sum_{j=1}^J v_{jr} y_j \right) w_{kr} \\
&= \sum_{r=1}^R (u_r^\top x) \cdot (v_r^\top y) w_{kr},
\end{aligned}$$

for every $k = 1, \dots, K$. Since this holds for every k , we can write

$$(4) \quad f(x, y) = \sum_{l=1}^r (u_l^\top x) \cdot (v_l^\top y) w_l.$$

Here we highlight with $(\cdot)$ the active multiplications. Now, evaluating f requires exactly R active multiplications (and $O(r)$ additions).

2.2 Recursive matrix multiplication

Recall that matrix multiplication for two matrices $A \in \mathbb{R}^{M \times N}$ and $B \in \mathbb{R}^{N \times P}$ is defined as

$$C = AB = \left(\sum_{k=1}^n A_{ik} B_{kj} \right)_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq p}} = [A_{:1} | \dots | A_{:n}] \begin{bmatrix} B_{1:} \\ \vdots \\ B_{n:} \end{bmatrix} = \sum_{k=1}^n A_{:k} B_{k:}.$$

From this expression easily follows the bilinearity of the map $(A, B) \mapsto AB$, and we indicate with $\langle M, N, P \rangle$ the form that represents the multiplication between A and B . By this property, we can rewrite Equation 2 putting $x = \text{vec}(A)$, $y = \text{vec}(B)$ and $C = \text{vec}(C^\top)$. This fixes the natural ordering on the entries of $\text{vec}(\mathbf{A})$ and $\text{vec}(\mathbf{B})$ and the inverse ordering on $\text{vec}(\mathbf{C}^\top)$. This yields a representation of the tensor $\mathcal{T} \in \mathbb{R}^{MN \times NP \times MP}$ of matrix multiplication $\langle M, N, P \rangle$ from Equation 2 such that

$$(5) \quad \text{vec}(\mathbf{C}^\top) = \mathcal{T} \times_1 \text{vec}(\mathbf{A})^\top \times_2 \text{vec}(\mathbf{B})^\top.$$

If we are given a CP decomposition of rank r of the tensor $\mathcal{T} = \llbracket U, V, W \rrbracket$, by Equation 1 this expression becomes

$$(6) \quad \text{vec}(\mathbf{C}^\top) = \sum_{l=1}^r \underbrace{(\mathbf{u}_l^\top \text{vec}(\mathbf{A}))}_{s_l} \cdot \underbrace{(\mathbf{v}_l^\top \text{vec}(\mathbf{B}))}_{t_l} \mathbf{w}_l = \sum_{l=1}^r \underbrace{(s_l \cdot t_l)}_{m_l} w_l,$$

The expressions above allow us to define fast matrix multiplications algorithms for any base case $\langle m, n, p \rangle$.

2.3 The classic matrix multiplication algorithm tensor formulation

We present the $\langle 2, 2, 2 \rangle$ classical matrix multiplication algorithm in the context of tensors. When $M, N, P > 2$ we can always partition them in blocks as:

$$(7) \quad \begin{matrix} \overbrace{[M/2]}^{[M/2]} \{ \overbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}^{\substack{[N/2] & [N/2]}}, \quad \overbrace{[N/2]}^{[N/2]} \{ \overbrace{\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}}^{\substack{[P/2] & [P/2]}} \end{matrix}$$

where the sizes of the subblocks are depicted in the above equation. This algorithmic formulation is often called *divide-and-conquer*, as it consists in breaking the problem into smaller subproblems, and later recombining the results. In particular, we can view the product AB as the product between 2×2 block matrices

$$(8) \quad \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \cdot \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

The naive algorithm proceeds by combining a set of eight matrix multiplications with four additions:

$$\begin{aligned} M_1 &= A_{11} \cdot B_{11} & M_2 &= A_{12} \cdot B_{21} & M_3 &= A_{11} \cdot B_{12} \\ M_4 &= A_{12} \cdot B_{22} & M_5 &= A_{21} \cdot B_{11} & M_6 &= A_{22} \cdot B_{21} \\ M_7 &= A_{21} \cdot B_{12} & M_8 &= A_{22} \cdot B_{22} \end{aligned}$$

$$\begin{aligned} C_{11} &= M_1 + M_2 & C_{12} &= M_3 + M_4 \\ C_{21} &= M_5 + M_6 & C_{22} &= M_7 + M_8 \end{aligned}$$

The number of flops performed by standard matrix multiplication for $N \times N$ matrices is $T(N) = 8T(\frac{N}{2}) + 4(\frac{N}{2})^2$ for $N > 1$. This can be solved by the master method to get $T(N) = 2N^3 - N^2$, where we assume that N is a power of two. In Section 4.1 we explain how to handle all dimensions.

For example, when $m = n = p = 2$, we get the tensor with frontal slices

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

This yields, for example,

$$X_3 \times_1 \text{vec}(A)^\top \times_2 \text{vec}(B)^\top = [a_{11} \ a_{21} \ a_{12} \ a_{22}] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{12} \\ b_{22} \end{bmatrix} = a_{21}b_{11} + a_{22}b_{21} = c_{21}.$$

Note that the formula in Equation 6 in the context of block matrix multiplication in Equation 8 yields to

$$\text{vec}(A) = \begin{bmatrix} A_{11} \\ A_{21} \\ A_{12} \\ A_{22} \end{bmatrix}, \quad \text{vec}(B) = \begin{bmatrix} B_{11} \\ B_{21} \\ B_{12} \\ B_{22} \end{bmatrix}, \quad \text{vec}(C^\top) = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{21} \\ C_{22} \end{bmatrix}.$$

Also the factors s_l and t_l in Equation 6 become

$$s_l = u_l^\top \text{vec}(A) = \sum_{i=1}^4 U_{li} \text{vec}(A)_i, \quad t_l = V_l^\top \text{vec}(B) = \sum_{i=1}^4 V_{li} \text{vec}(B)_i.$$

2.4 Strassen algorithm using CP

The best-known fast algorithm is due to Strassen, and follows the same $\langle 2, 2, 2 \rangle$ block structure:

$$\begin{aligned} S_1 &= A_{11} + A_{22} & S_2 &= A_{21} + A_{22} & S_3 &= A_{11} \\ S_4 &= A_{22} & S_5 &= A_{11} + A_{12} & S_6 &= A_{21} - A_{11} \\ S_7 &= A_{12} - A_{22} \\ T_1 &= B_{11} + B_{22} & T_2 &= B_{11} & T_3 &= B_{12} - B_{22} \\ T_4 &= B_{21} - B_{11} & T_5 &= B_{22} & T_6 &= B_{11} + B_{12} \\ T_7 &= B_{21} + B_{22} \end{aligned}$$

$$\begin{aligned} M_r &= S_r T_r, \quad 1 \leq r \leq 7 \\ C_{11} &= M_1 + M_4 - M_5 + M_7 & C_{12} &= M_3 + M_5 \\ C_{21} &= M_2 + M_4 & C_{22} &= M_1 - M_2 + M_3 + M_6 \end{aligned}$$

This algorithm uses 7 matrix multiplications and 18 matrix additions. The number of flops performed by the algorithm is $T(N) = 7T(\frac{N}{2}) + 18(\frac{N}{2})^2$ for $N > 1$, solving the recursion the complexity is $T(N) = 7N^{\log_2 7} - 6N^2 = O(N^{\log_2 7}) = O(N^{2.81})$. This algorithm corresponds to a low-rank tensor decomposition represented by the following triplet of matrices, each with 7 columns:

$$\begin{aligned} U &= \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & -1 \end{bmatrix}, \\ V &= \begin{bmatrix} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & 1 & 0 & 1 \end{bmatrix}, \\ W &= \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Table 2: Summary of fast algorithms. Algorithms without citation were found by the authors of [1]. The number of multiplications is equal to the rank R of the corresponding tensor decomposition. The multiplication speedup per recursive step is the expected speedup if matrix additions were free. Note that this speedup does not determine the fastest algorithm because the maximum number of recursive steps depend on the size of the subproblems created by the algorithm.

Algorithm base case	Number of multiplies (fast)	Number of multiplies (classical)	Multiplication speedup per recursive step
$\langle 2, 2, 3 \rangle$	11	12	9%
$\langle 2, 2, 5 \rangle$	18	20	11%
$\langle 2, 2, 2 \rangle$ [5]	7	8	14%
$\langle 2, 2, 4 \rangle$	14	16	14%
$\langle 3, 3, 3 \rangle$	23	26	17%
$\langle 2, 3, 3 \rangle$	15	18	20%
$\langle 2, 3, 4 \rangle$	20	24	20%
$\langle 2, 4, 4 \rangle$	26	32	23%
$\langle 3, 3, 4 \rangle$	29	36	24%
$\langle 3, 4, 4 \rangle$	38	48	26%
$\langle 3, 3, 6 \rangle$ [6]	40	54	35%

While we cannot use less than 7 multiplications for the $\langle 2, 2, 2 \rangle$ algorithm [4], there are natural extensions to Strassen algorithm:

1. We can reduce the number of addition from 18 down to 15, which leads to Strassen-Winograd algorithm .
2. We can change constant on the leading term by choosing a bigger base case dimension (and using the standard algorithm (also called *gemm*) on the base case).
3. We can use blocking schemes different from $\langle 2, 2, 2 \rangle$.

cite

We implement this extension in practice using the Rust programming language, targeting exact algorithms arising from the CP decomposition matrices made freely available by Benson and Ballard [1]. Table 2 shows a summary of fast algorithms for different base cases.

3 Project overview

3.1 Fork overview and comparison with c++ version

The implementation consists of a Rust port of the c++ project made by Benson and Ballard summarized in [1], available in the relative repository under BSD 2-Clause. We will refer to this project as "the C++ project" or "Benson and Ballard's project". They provide a fast and efficient framework for benchmarking parallel and sequential fast matrix multiplication algorithms, achieved through code generation, using lapack and MKL dgemm for numerical computation, and OpenMP for parallelization. We develop a Rust project that aims to replicate a parallel framework for fast multiplication algorithms, but with structural

differences and current limitations. The main similarities are that both use MKL for base, but Rust project also uses Faer. In the C++ project, MKL is imported, while in Rust it wrapped via FFI. Besides tool choices, the C++ algorithms are code-generated by a script that parses CP tensor decomposition slices and translates the matrix multiplication expression into C++ code, while Rust loads the tensor at runtime¹ and then applies the pseudocode in Algorithm 1. The advantage of the first is speed, at the cost of more complex code and parallelism. The advantage of the second is simplicity of code, at the cost of speed (at the current state, but which it could be improved by adding optimization, we will address these issues in Section 6). The C++ project can handle multiplications between rectangular matrices, as well as APA algorithms, while our does not. Also, while it could work, we did not tested subexpression eliminated algorithms.

3.2 Algorithm schema

We now discuss the algorithm schema for Strassen algorithm for clarity, though results generalize easily for any $\langle m, n, p \rangle$ base case. Algorithm 1 shows our implementation of Strassen’s algorithm, which easily extends to other CP algorithms by changing base conditions and block sizes—this is how it is implemented in practice.

Algorithm 1 Strassen algorithm using CP factorization.

```

1: procedure STRASSEN( $A, B, N$ )
2:   if STOP-RECURSING is True then
3:     return  $A \cdot B$ 
4:   end if
5:   if  $N = 2$  then
6:     STRASSEN-BASE-MATMUL( $A, B$ )
7:   end if
8:   if  $N \bmod 2 > 0$  then
9:     DYNAMIC-PEELING( $A, B, N$ ) ▷ Recursive call on peeled blocks.
10:  end if
11:  Compute  $\text{vec}(A)$  and  $\text{vec}(B)$  with block size  $\frac{N}{2}$ .
12:  for  $r = 1, \dots, R$  do
13:     $S_r \leftarrow \sum_{i=1}^4 U_{i,r} \text{vec}(A)_i$ 
14:     $T_r \leftarrow \sum_{i=1}^4 V_{i,r} \text{vec}(B)_i$ 
15:     $M_r \leftarrow \text{STRASSEN}(S_r, T_r, \frac{N}{2})$  ▷ Recursive call.
16:  end for
17:   $\text{vec}(C^\top) = \sum_{r=1}^R M_r w_r$ 
18:  return  $C$ .
19: end procedure

```

¹when program is launched, before doing the multiplication, which does not affect speed since the matrices decomposition is cached at runtime

3.3 Execution platform and reproducibility

Platform

Since the comparison targets both sequential and parallel matrix multiplication kernels, we ran all experiments on one node of a dual-socket AMD EPYC 7763 server (64 physical cores per socket, 2 threads per core) with 2 TB of RAM. The system ran Ubuntu 24.04.5 LTS with Linux 5.15.0-179-generic on x86_64. For sequential benchmarks, library-level threading was restricted with `OMP_NUM_THREADS=1` and `MKL_NUM_THREADS=1`. These choices remove scheduler migration and library-level parallelism from the measurements to isolate microarchitectural performance and compiler optimizations without the interference of thread scheduling and synchronization overheads.

Environment

The software environment is configured and reproduced using Environment Modules. The configuration loads `gcc/13.2.0` as the base compiler toolchain, alongside the `intel-oneapi-compilers` version `2025.0.4-gcc-11.4.0`. To provide an optimized BLAS/LAPACK backend, the environment loads `intel-oneapi-mkl`. We are aware that this makes the environment not consistent, but it was not possible to match the correct versions without incurring compilation errors.

Compilation

When building the C wrappers and Rust bindings, all compilation targets are explicitly optimized for the server target architecture. In the C++ project, to support both sequential and parallel modes within a single executable, the MKL backend was dynamically linked against the LLVM OpenMP threading layer passing `-liomp5`. We build the Rust code with the `nightly-2026-06-23` toolchain because this release uses LLVM 22, which generates optimized vector instructions leveraging AVX2 and FMA. Since the C/C++ wrappers and Rust code compile with matching CPU targets, a custom `build.rs` script compiles the Intel MKL FFI C wrapper (`mkl.c`) using `gcc`. During linking, `build.rs` ensures that the Intel MKL dynamic libraries are correctly loaded. To eliminate the need for manual `LD_LIBRARY_PATH` configuration at runtime, the MKL library path is compiled into the executable binaries as a runtime.

4 Practical optimizations

Fast matrix multiplication algorithms such as Strassen face two significant problems in practical implementation, which are recursion cutoff and input matrices sizes not matching with base case. In the parallel case, there are also scheduling problems to consider to avoid algorithm overhead.

4.1 Dynamic Peeling

In fast matrix multiplication algorithms we want to multiply $M \times N \times P$ matrices by splitting them into four blocks as in Equation 7. If M , N and P are not divisible by the base case $\langle m, n, p \rangle$, then the matrices cannot be partitioned into equal blocks. The easiest fix for this is padding the matrix with zeros until an

even size is reached, but this adds a significant memory waste. A more efficient approach, called *dynamic peeling*, consists in stripping the extra row off and/or column and adding their contribution to the final results.

We present dynamic peeling for Strassen’s algorithm for the sake of clarity—the construction easily adapts to other base cases. For the matrix multiplication $C = AB$, where A is an $M \times N$ matrix and B is an $N \times P$ matrix. The matrix A is split into an $(M - 1) \times (N - 1)$ core matrix A_{11} , alongside its peeled boundary vectors a_{12} , a_{21} , and the scalar corner element a_{22} . Similarly, the matrix B is split into an $(N - 1) \times (P - 1)$ core matrix B_{11} , with its corresponding peeled components b_{12} , b_{21} , and b_{22} :

$$A = \left[\begin{array}{c|c} \overbrace{A_{11}}^{n-1} & \overbrace{a_{12}}^1 \\ \hline a_{21} & a_{22} \end{array} \right] \left. \vphantom{\begin{array}{c|c} \overbrace{A_{11}}^{n-1} & \overbrace{a_{12}}^1 \\ \hline a_{21} & a_{22} \end{array}} \right\}^{m-1}_1, \quad B = \left[\begin{array}{c|c} \overbrace{B_{11}}^{p-1} & \overbrace{b_{12}}^1 \\ \hline b_{21} & b_{22} \end{array} \right] \left. \vphantom{\begin{array}{c|c} \overbrace{B_{11}}^{p-1} & \overbrace{b_{12}}^1 \\ \hline b_{21} & b_{22} \end{array}} \right\}^{n-1}_1$$

The final block matrix product is computed directly through a combination of the core Strassen multiplication ($A_{11}B_{11}$) and low-rank vector/scalar fixup modifications:

$$\left[\begin{array}{c|c} \overbrace{C_{11}}^{p-1} & \overbrace{c_{12}}^1 \\ \hline c_{21} & c_{22} \end{array} \right] \left. \vphantom{\begin{array}{c|c} \overbrace{C_{11}}^{p-1} & \overbrace{c_{12}}^1 \\ \hline c_{21} & c_{22} \end{array}} \right\}^{m-1}_1 = \left[\begin{array}{c|c} A_{11}B_{11} + a_{12}b_{21} & A_{11}b_{12} + a_{12}b_{22} \\ \hline a_{21}B_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{array} \right]$$

Here, only the sub-product $A_{11}B_{11}$ is executed recursively using Strassen’s matrix multiplication, while all other terms are computed via standard matrix multiplication.

4.2 Parallel algorithms for shared memory

Matrix multiplication is a problem subject to both computational and memory bound. The first is limited by the amount of computation the device can handle (i.e cores,cpu/gpu speed), while the latter by physical storage of the matrices (i.e available RAM memory). In a parallel implementation, when dealing with recursive algorithms which use divide-et-impera (like Strassen), it is important in practice to balance algorithmic load between threads to avoid overhead caused by both taking recursive branches and doing parallel matrix multiplication. To address this issue, we adopt the scheduling schema suggested by Benson and Ballard in their work [1]. We use Rust’s Rayon to handle parallelization, and we can summarize the parallel scheduling as following:

- **BFS:** Each recursive matrix multiplication routine and the associated matrix additions are launched by Rayon as an iterable, and then are collected in an array of matrices $[M_1, \dots, M_r]$.
- **DFS:** Each vendor matrix multiplication call uses all thread, and matrix multiplications are always fully parallelized.
- **Hybrid:** The Hybrid approach developed in [1] compensates for the load imbalance in BFS by applying DFS on a subset of the base case problems. With L levels of recursion and P threads, the hybrid algorithm applies task parallelism (BFS) to the first $R^L - (R^L \bmod P)$ multiplication.

Because the number of subproblems is a multiple of P , this part is load balanced. On the remaining $R^L \bmod P$ subproblems, all threads are used on each matrix multiplication (DFS).

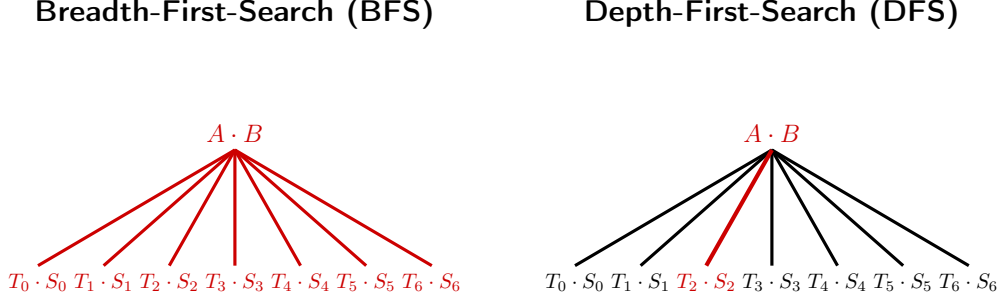


Figure 1: BFS and DFS load balancing. The parallelized tasks are in red.

4.3 Recursive cutoff strategies

In divide-and-conquer algorithms such as Strassen, we cannot afford taking many recursive steps, as spawning many children constitutes a significant recursive overhead, especially for larger matrices. Thus, finding an optimal cutoff is important for stopping the recursion and switch to standard GEMM. We try two techniques: the first is breaking by size, that is when the sizes of matrices in the recursion fall under a certain size, the latter is taking a fixed number L of recursions.

The authors in [1] explain a heuristic process for finding the optimal cutoff size, which can be resumed in looking at the performance plot of the GEMM function, where (figure) the function exhibits a ramp-up and then flatten for large dimensions. They take a recursion step only if the recursive sub-problem fall in the flat part of the curve. This is done manually by the authors. During the experimentation we developed a function to plot this curve, and its derivative (using splines) to automatically find this size. In practice, we never use this as we believe that our code is not efficient enough to implement this strategy.

5 Experimental results

5.1 Metrics

All benchmarks are recorded using Rust’s system clock, performing warm-up before doing measuring each matrix multiplication. In order to compare the performance of matrix multiplication algorithms with different computational costs, we use the *effective* GFLOPS metric for $M \times N \times P$ matrix multiplication:

$$(9) \quad \text{effective GFLOPS} = \frac{2MNP - MP}{\text{time in seconds} \cdot 10^9},$$

where the numerator can be rewritten as $2MNP - MP = MNP + M(N - 1)P$, which is the number of multiplications and additions performed for classical matrix multiplication. This allows us to compare all of the algorithms on an inverse-time scale, normalized by problem size [1]. For fast algorithms, considering both multiplications and additions in the metric is more accurate than taking multiplications only [7].

5.2 Measuring impact of base GEMM

We aim to understand how the choice of the base matrix multiplication impacts performance of fast algorithms. In our experiments, we use random square matrices with 64bit floating-point entries in $(-1, 1)$ of sizes $N \in \{2, 4, \dots, 2^k\}$. As the standard matrix multiplication algorithm implemented in GEMM costs $O(n^3)$ while Strassen costs $O(n^{2.81})$, we expect it to beat GEMM² for a large N , that is, we are interested in finding the N such that using a fast matrix algorithm is faster than doing standard matrix multiplication. Such N cannot be determined a priori, as it depends on factors such as platform and implementation. In other words, we expect fast algorithms to surpass the curves in Figure 2.

5.3 Comparison of MKL dgemm FFI and Faer matmul

In this work, we consider two different matrix multiplication routines: MKL dgemm and faer base matmul.

- **Faer matmul:** The method is entirely written in Rust and is part of the faer library, and it wants to be a small performant matrix multiplication function, with fewer microkernel optimization than standard GEMMs. See [8] for a method overview from the library’s author.
- **MKL dgemm:** is a routine in the Intel Math Kernel Library (MKL) for multiplying double-precision matrices. In Rust, this routine runs via a Foreign Function Interface (FFI) calling the C wrapper `cbblas_dgemm`. In the C++ project, the function is called directly. Both projects use the same library version on the architecture.

From Figure 2, we see that all functions have a ramp-up phase and then flatten for larger sizes—even if the parallel case is not flattening yet, we expect it to flatten after few more runs. Also, dgemm performs better than faer for larger sizes, and we believe this is due to the many low-level optimizations in dgemm. Further, there is a discrepancy between Rust FFI dgemm and the C++ code which tends to decrease for larger size. We think that this might be due to the type conversion which is performed during the binding.

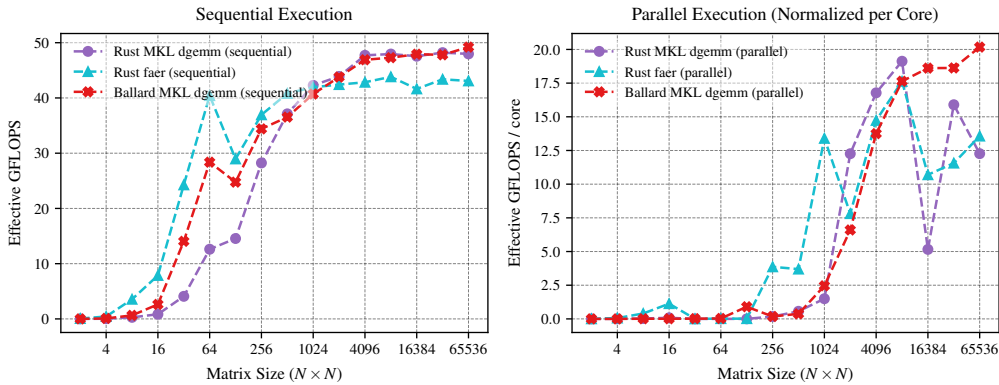


Figure 2: Comparison between MKL and faer in sequential and parallel execution.

²which is in practice more optimized and a linear algorithm

5.4 Comparison of the level and size recursive cutoff strategies

The experimental results for sequential and parallel cutoff strategies are shown in Figures 3a and 3b, respectively. Figures 4a and 4b show comparison of Strassen and base GEMM for the parallel case. In general, we notice that while base matmul performance tends to be different between MKL and faer, the respective Strassen variants tend to have similar performance, as Figures 3a and 3b show. In the sequential case in Figure 3a, the best performance is achieved when cutoff is 2048 or when two levels of recursion are taken. We notice that in all cases Strassen outperforms GEMM, and growth is expected to continue for larger sizes, though we didn't test it as it would have required high computational time. In the parallel case, Figure shows that for optimal recursive levels and cutoffs, for example when Level = 2 or for Cutoffs 8129 or 16384. Compared to the MKL base case in Figure 4a and Figure 4b Strassen, especially the BFS and Hybrid variants perform better, showing peak performance with 2 or 3 levels of recursion and for greater cutoffs (when ≥ 4096). .

improve
with
dgemm
better data

5.5 Finding the optimal cutoff for the parallel case

Figure 5 shows different recursive size thresholds for the parallel case, where BFS outperforms DFS in both variants, seeming also not sensible to the base case as DFS. Compared to the sequential case in Figure 3a, optimal performance is found for larger cutoff size, such as when ≥ 8102 . We think that this is due that smaller cases trigger more recursive steps, causing more overhead. In Figure 5b, the faer variant shows

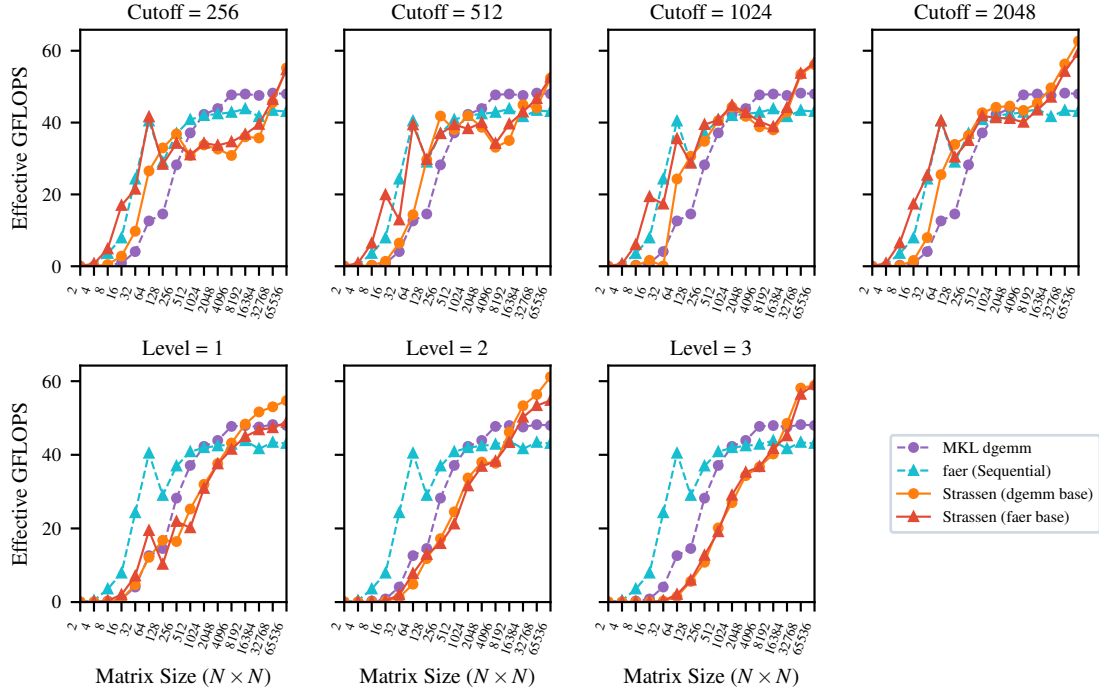
5.6 Numerical comparison with Benson&Ballard Strassen

We perform a numerical performance comparison between our optimized Rust implementation of Strassen's algorithm and the reference C++ implementation by Benson and Ballard [1]. The results are shown in Figure 6. We compare the algorithms across one, two, and three levels of recursion using different thread scheduling paradigms: Sequential, Parallel DFS, Parallel BFS, and Parallel Hybrid. We observe that our implementation is competitive in the sequential case, where the dgemm-backed version yielding the highest performance. In the DFS case, Ballard code outperforms our algorithm, we think this is due to their better parallelization of additions.

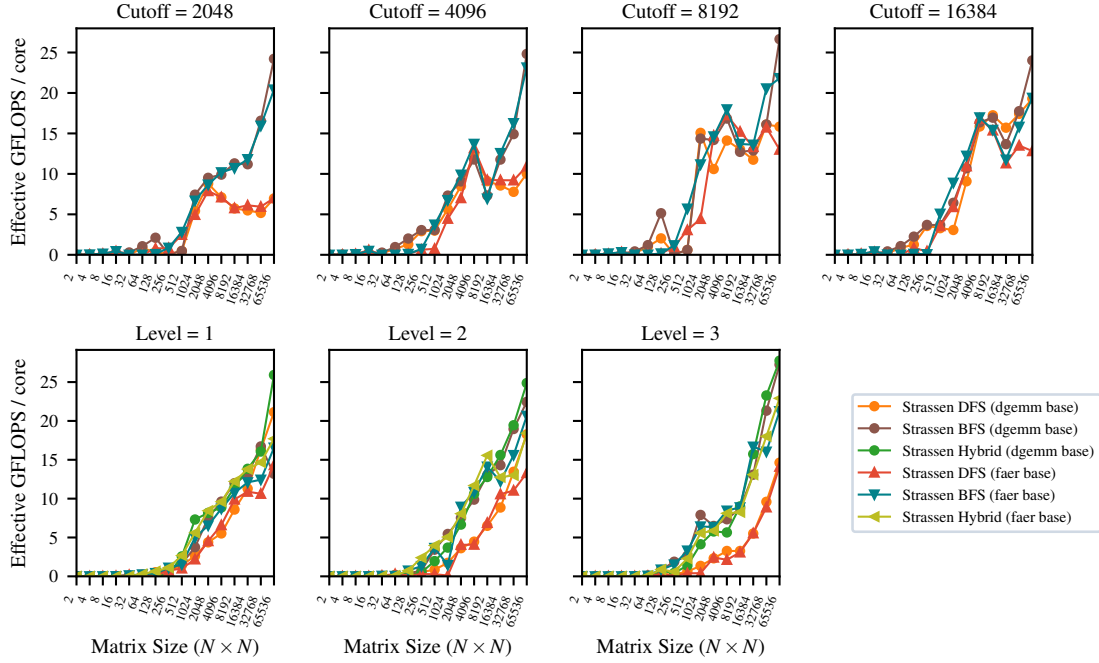
6 Conclusion & Further work

In conclusion, we noticed fast matrix multiplication algorithm show more optimal performance than the base GEMM libraries both in sequential and parallel cases, even with little performance optimizations. We believe that results of our work can be further improved by pursuing some of the following steps:

- **Profiling and SIMD** We believe that code can be made more efficient by doing a more fine refactory with profiling result. Many algorithm steps can be optimizing by introducing more SIMD operations.
- **GEMM comparison:** Though we choose mkl-dgemm for a better comparison, our ffi bindings easily extend to any other GEMM, such as OpenBlas, which we leave to further work.



(a) Sequential execution.



(b) Parallel execution (normalized per core).

Figure 3: Performance comparison of Strassen variants (DFS, BFS, Hybrid) across recursion levels and size cutoffs in sequential and parallel executions.

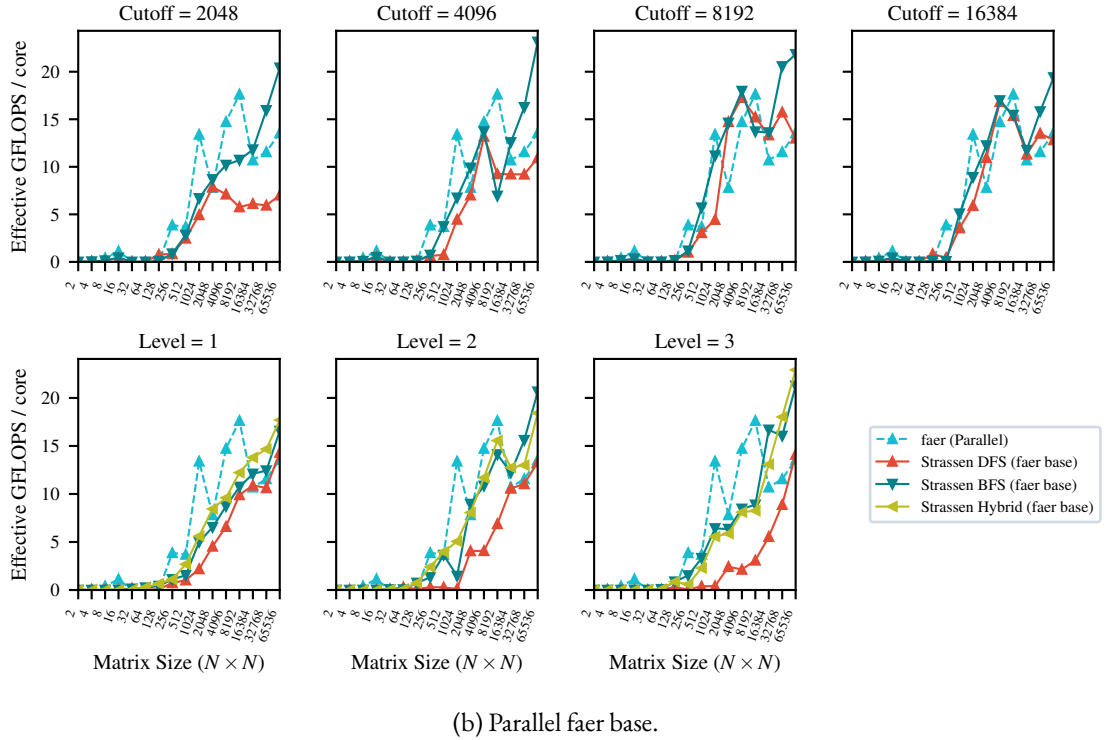
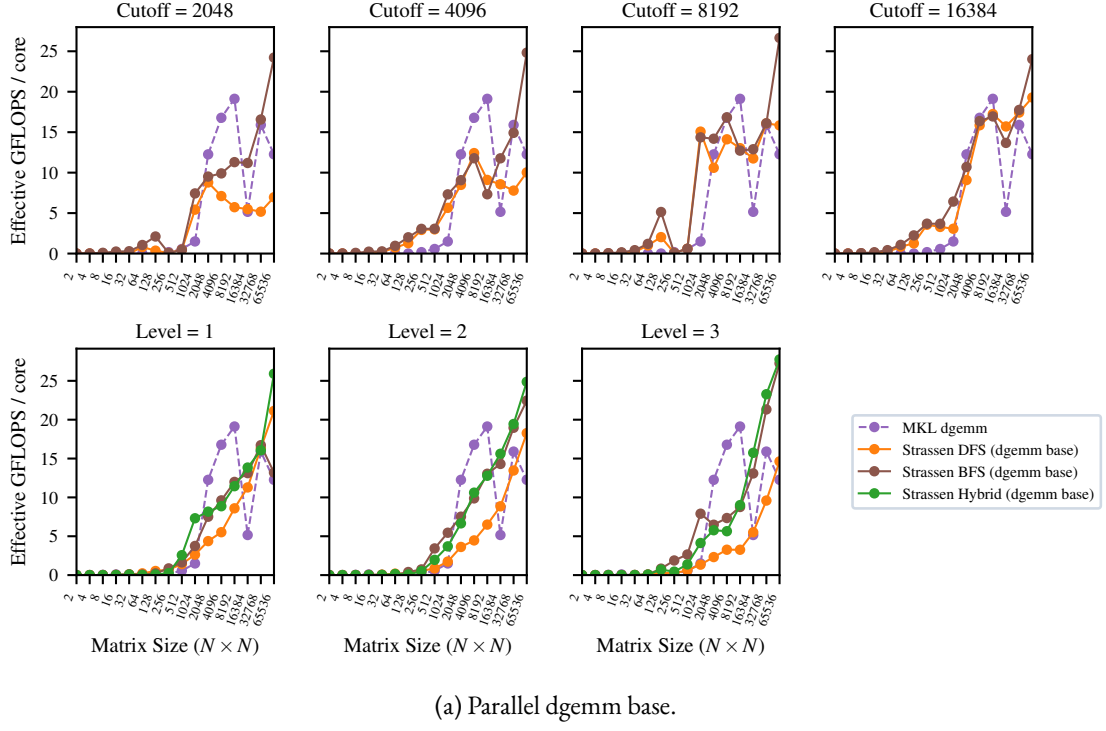


Figure 4: Parallel performance comparison (normalized per core) of Strassen variants (DFS, BFS, Hybrid) against parallel GEMM, across cutoff sizes (first row), and recursion levels (second row).

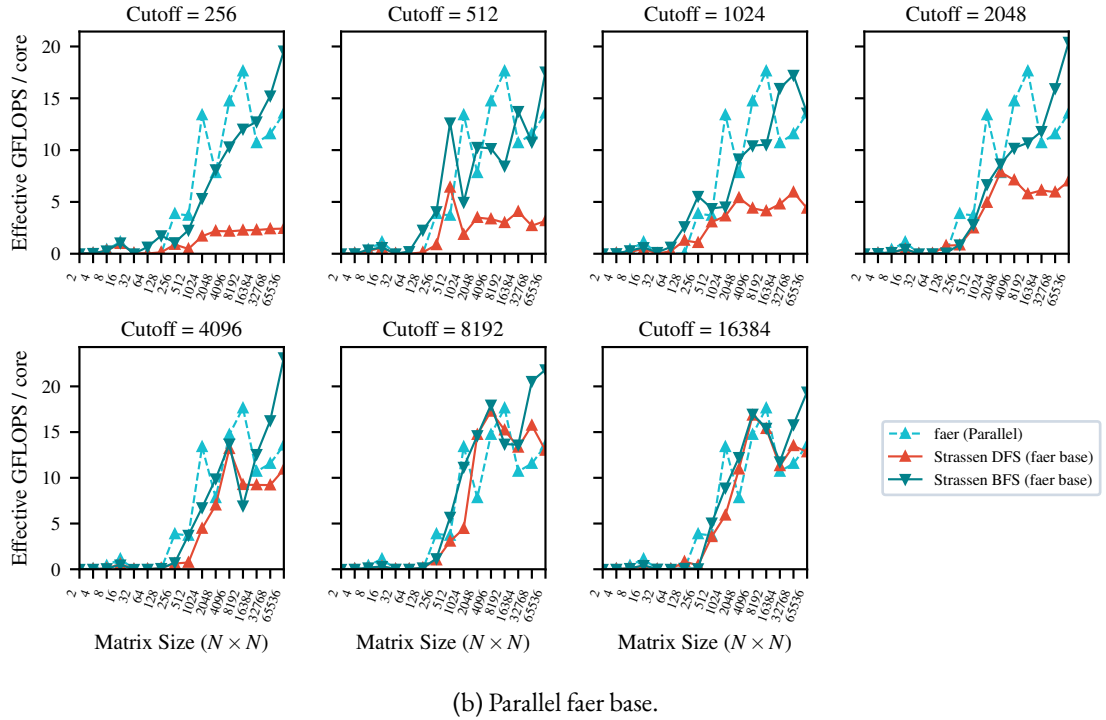
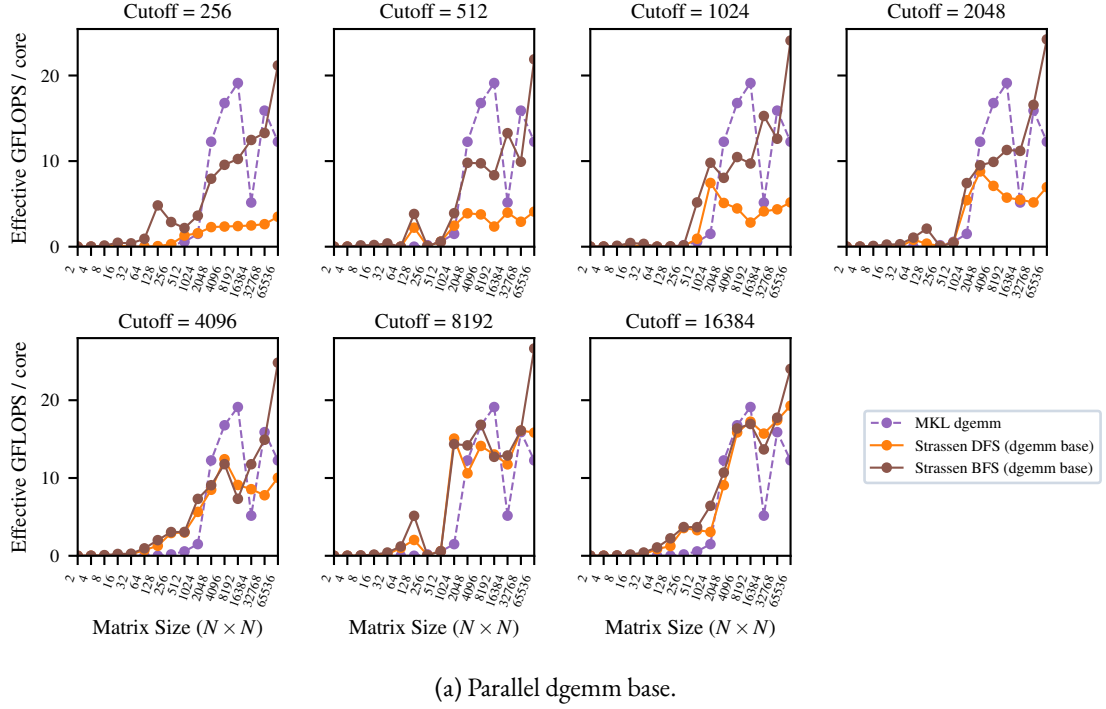


Figure 5: Parallel matrix multiplication performance comparison (normalized per core) of Strassen variants (DFS, BFS, Hybrid) against parallel GEMM across size cutoffs.

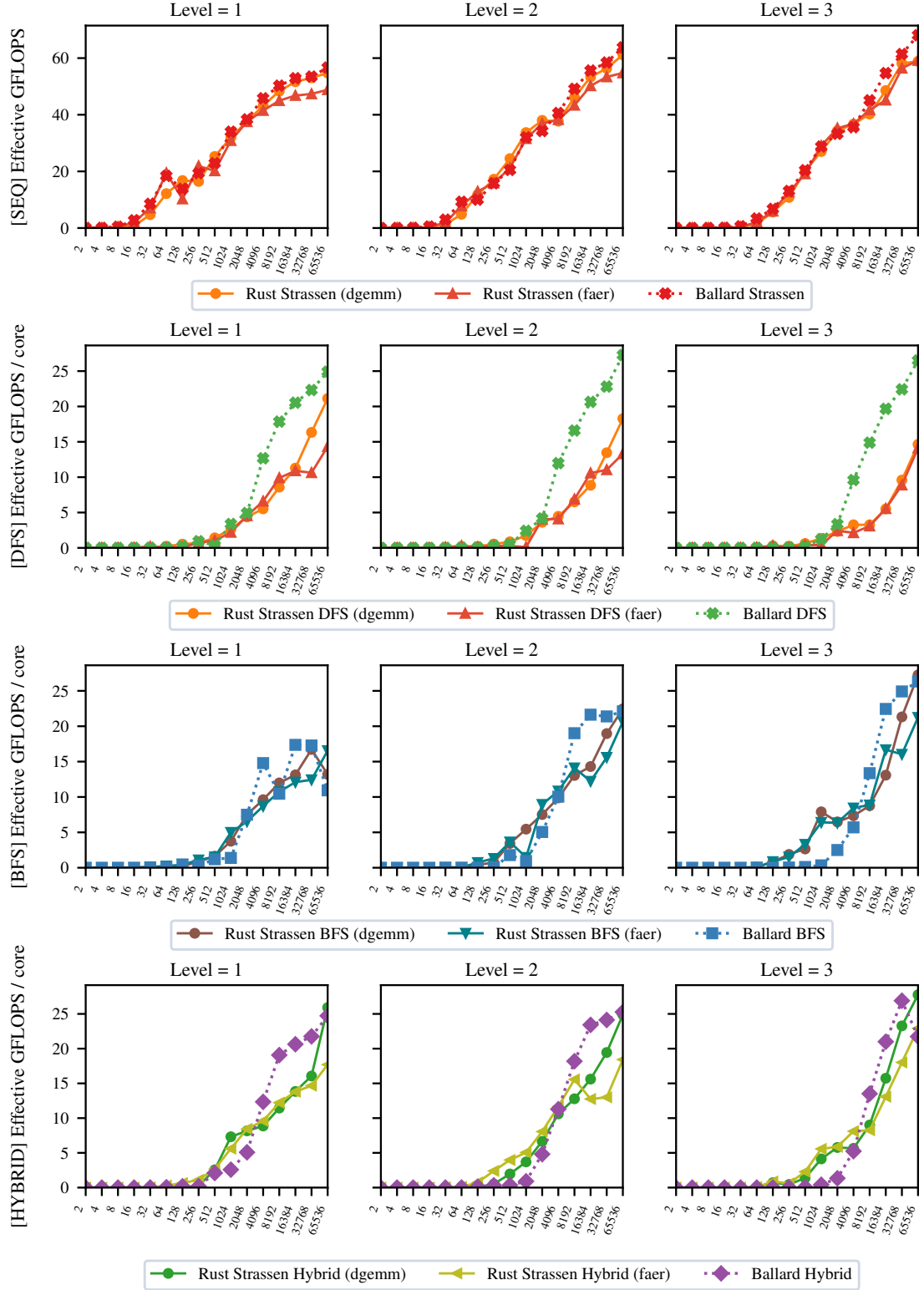


Figure 6: Performance comparison of our Rust Strassen implementation (using dgemm and faer base cases) against the Benson and Ballard reference benchmarks, sequential and parallel execution.

- **Gpu/distributed version:** We would like to port the project to GPU or distributed computing to extend results with higher parallelism and test larger sizes.
- **Run on Xeon:** The project could be run on high performance Intel Xeon nodes on cluster, which might show impact of Intel MKL dgemm proprietary optimizations.

7 Work reproducibility

All the code is available at the repository [fast-matmul](#).

8 Acknowledgments

We acknowledge the authors of [2] for providing instructions for linking MKL with Rust, and the department of Mathematics of the University of Pisa for offering access to their HPC cluster.

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